

Yerrabandla Sai Pranav Reddy

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LinkedIn

GitHub

Portfolio

Education

Neil Gogte Institute of Technology

2023 – Present

Bachelor of Engineering in Computer Science and Engineering — CGPA: 9.004

Hyderabad, Telangana

Experience

DrugParadigm

April 2025 – Present

AI Research Intern, Onsite

Hyderabad, Telangana

- Architected and deployed structure-aware generative and regression models for peptide and enzyme modalities using Graph Neural Networks (GNNs), Geometric Deep Learning, and Diffusion models in PyTorch. Engineered end-to-end training and inference pipelines on NVIDIA DGX multi-GPU systems, optimizing memory utilization to handle large-scale biological datasets.
- InversePep – Peptide Inverse Folding (Generative Modeling):** Developed a structure-conditioned generative model for peptide inverse folding, trained on a massive dataset of **38,000+ sequences** and structures. Integrated Transformer encoders, conditional Diffusion models, and GVP-based Graph Neural Networks (GVP-GNN). Achieved a **0.51 TM-score**, a **18%+** improvement over ProteinMPNN(0.31) and ESM-IF1(0.33) benchmarks..
- GraphEnzyme – Enzyme Kinetics Prediction (Regression Modeling):** Designed a multimodal framework for enzyme kcat prediction using **23,000+ sequences**, structures, and SMILES strings. Fused GVP-GNNs for structures, GCNs for substrate molecules, and ChemBERTa for SMILES with Squeeze-and-Excitation (SE) blocks for adaptive weighting. Achieved a **PCC of 0.81** and **R-Squared of 0.64**, significantly improving **5%+** predictive consistency via dataset expansion and architectural tuning.

Projects

NyayLens - Transformer-Based Legal QA System

- Architected an end-to-end Legal AI Research Platform over 26K+ Supreme Court judgments (**298K+ embeddings**), leveraging **Legal-BERT + Legal-PEGASUS** to deliver citation-aware answers **90%** faster than manual review.
- Engineered a **FastAPI Agentic Orchestrator** dynamically dispatching across Document-QA (30K-char context window), **Hybrid Retrieval (FAISS + SQLite FTS5)**, and Legal Summarization - outperforming vanilla vector-search baselines on precision and factual consistency.
- Delivered production-grade full-stack system with **React + custom HuggingFace Transformer architectures** for privacy-centric local summarization and **Llama 3.1 8B (Groq)** for sub-second conversational inference, with power-user slash-commands (/summarize, /decision) for complex multi-document workflows.

PranaSetu – Multimodal Mental Health Risk Assessment Platform

GitHub

- Developed a production-grade multimodal monitoring system integrating BERT-based NLP and EfficientNet-based Computer Vision to perform real-time risk assessments from heterogeneous social and user-submitted data.
- Engineered a scalable full-stack architecture utilizing Node.js, Express, and MongoDB, with a high-performance FastAPI/PyTorch backend integrating Gemini LLM for automated clinical interpretation and longitudinal trend tracking.

Bloom - AI-Powered Plant Disease Detection

Live Demo | GitHub

- Built an end-to-end plant disease classification system on **87k+ images** across **38 classes** to enable automated crop health diagnosis. Implemented CNN architectures from scratch in NumPy and production-grade TensorFlow models, achieving **92% accuracy** with custom CNN and **97% accuracy** using optimized TensorFlow pipelines.
- Deployed a production-ready system with frontend (Vercel), backend APIs (Render), MongoDB for user and prediction management and model hosting on HuggingFace for real-time inference.

Technical Skills

- Programming:** Python, Java, JavaScript, C
- Core CS:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks
- Web Development:** Node.js, Express.js, REST APIs, React.js, MongoDB, PostgreSQL, HTML, CSS
- Machine Learning & AI:** PyTorch, TensorFlow, Neural Networks (ANNs CNNs), Transformers (BERT, T5), Graph Neural Networks (GNNs GCNs GVP), Diffusion Models
- Tools & Systems:** Git, Postman, VS Code

Achievements and Certifications

- Ranked in the **Top 5%** among 50,000+ participants in the Amazon ML Challenge (2025).
- Secured **3rd place** in a 12-hour internal hackathon at Keshav Memorial Engineering College (KMEC).
- Oracle Certified: Oracle Cloud Infrastructure 2025 AI Foundations Associate.
- IBM Certified Data Analyst – IBM Professional Certification.